

COLLISION-MARKED TRAJECTORY CLUSTERS AND JOINT DYNAMIC LARGE DEVIATIONS FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. We construct the compatible space–time Green trace graph for deterministic hard spheres and control surplus contacts through the actual linearized collision map. Fixed-genealogy analytic transversality is summed by dominated convergence against factorial trajectory weights, avoiding a false genealogy-uniform exponent. Ovsyannikov source sewing, a continuous kinetic Hodge repair, and bounded-source exhaustion give the full density–actual-contact LDP and its microcanonical transfer.

Platform identifier: HSBG-LI0-v4. **Controlling revision:** ROUND12-REFEREE-POSITIVE-CLOSURE.
Registered proof source: B2_SUMMED_JACOBI_TRACE_LDP.tex.

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1. ROUND-TWELVE POSITIVE CLOSURE: EXACT TRACE GRAPHS, SUMMED JACOBI TRANSVERSALITY, AND GLOBAL MARKED LDP

The first-surplus estimate is derived from the actual linearized hard-sphere collision map. No rigid subtree translation, QR reset, or genealogy-uniform Łojasiewicz exponent is used. Fixed-genealogy transversality is summed by dominated convergence against the factorial cluster weights.

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 60F10, 82C40; Secondary 76P05, 82C22.

Key words and phrases. hard spheres, collision flow, trajectory clusters, recollisions, dynamic large deviations, Boltzmann–Grad limit.

1.1. The space–time trace graph. For every hierarchy level let $\mathcal{J}_k = (g_k, V_k g_k)$ be the space–time divergence-measure current on $[0, T] \times \Omega_k^\varepsilon$. Its normal trace on a collision face is the signed flux measure $\gamma_{k,ij}$ itself.

Theorem 1.1 (Closed compatible trace generator). *The graph consisting of $(g_k, \operatorname{div} \mathcal{J}_k, \gamma_{k,ij})_{k \geq 0}$, with incoming/outgoing traces related by elastic reflection, is closed in the weighted weak topology. For every graph test ϕ ,*

$$\int \phi d(\operatorname{div} \mathcal{J}_k) + \int \mathcal{J}_k \cdot d(D\phi) = \sum_{i < j} \int_{\Gamma_{ij}} \phi d\gamma_{k,ij} + \text{time-endpoint traces}.$$

The triangular free-flight/creation operator generates an Ovsyannikov evolution on radii $\alpha(t) = \alpha_0 - Ct$ and preserves the trace graph.

Proof. The Gauss–Green theorem for divergence-measure fields defines the normal trace as one measure, so no second velocity factor occurs. Reflection is a measure-preserving involution on incoming and outgoing faces. Uniform total variation and divergence bounds give weak compactness; testing the Green identity and reflection relation passes them to the limit, including endpoint traces. Collision-face intersections have zero flux under the weighted graph norm and are included through their oriented iterated traces. The creation operator loses Cdt of label radius on a time interval dt ; Picard iteration on the decreasing radius scale gives the stated evolution and graph preservation. $\square$

1.2. The actual Jacobi map. Fix a connected creation tree and its first surplus contact. Cut the tree at the youngest edge separating the two surplus labels. Use two chronological root variables before that edge and solve the creation constraints by the implicit function theorem. Let J_λ be the resulting two-dimensional transverse derivative of the surplus relative position. It includes normal variation, reflected-velocity derivatives, flight-time derivatives, and every intermediate collision.

Lemma 1.2 (Fixed-genealogy physical transversality). *For every fixed regular genealogy λ , $\det J_\lambda$ is a nonzero real-analytic function of the chronological variables off grazing and multiple-collision strata. Consequently there are $\beta_\lambda > 0$ and $C_\lambda < \infty$ such that*

$$\mathfrak{m}_\lambda\{s_{\min}(J_\lambda) < \rho\} \leq C_\lambda \rho^{\beta_\lambda}$$

after integration with the actual incoming flux weights.

Proof. Differentiate free flight and the reflection law $v' = v - ((v - v_*) \cdot \omega)\omega$ without freezing ω . The resulting Jacobi recursion is analytic on every homogeneous chronology. To show that a minor is not identically zero, choose the two root positions so that the two subtrees approach the youngest connecting collision with transverse relative translations. The implicit creation equations have full rank there, and the symplectic collision derivative sends the two translations to independent transverse Jacobi fields at the surplus time. Analytic sublevel estimates on the compact homogeneous chart, followed by

integration of the local inverse cosine against incoming flux, give the claim. The constants may depend on the fixed genealogy; no uniform exponent is asserted. $\square$

Let $a_m(T)$ be the total absolute cluster weight of all m -label connected genealogies. The tree estimate gives $\sum_m a_m(T) < \infty$ for $T \leq T_*$.

Theorem 1.3 (Summed first-surplus modulus). *There is a deterministic modulus $\omega_T(\varepsilon) \downarrow 0$ such that the total absolute weight of every connected cluster containing a surplus contact is at most*

$$h \omega_T(\varepsilon)$$

on a time block of length h . The estimate is uniform on every bounded particle/contact source ball and after every fixed number of source derivatives.

Proof. For a fixed genealogy, coarea on the two transverse variables and Lemma 1.2 show that its surplus-contact weight tends to zero as $\varepsilon \rightarrow 0$. It is bounded by the ordinary absolute cluster weight. Summing genealogies and using $\sum_m a_m(T) < \infty$, dominated convergence gives a modulus independent of genealogy depth. The first surplus time is integrated over the block, yielding the factor h . Bounded source marks only change the summable majorant. $\square$

1.3. Source sewing and grand-canonical pressure.

Theorem 1.4 (Partition-stable marked source sewing). *For every partition π of $[0, T]$ with mesh $|\pi|$ and every bounded predictable source, the block product differs from the exact marked trajectory evolution by*

$$C_T(\omega_T(\varepsilon) + |\pi|).$$

The analytic radius at time t is $\alpha_0 - Ct$, independent of the number of blocks. Two arbitrary partition sequences with mesh tending to zero have the same limit.

Proof. The noncyclic boundary error on a block is $O(h^2)$ and the cyclic sector is $h\omega_T(\varepsilon)$ by Theorem 1.3. Summing blocks gives $T\omega_T(\varepsilon) + CT|\pi|$. The generator estimate loses Ch of label radius, so losses add to CT rather than one fixed amount per block. For two partitions use their union; its mesh is the larger of the two meshes and satisfies the same estimate. This proves partition independence. $\square$

Theorem 1.5 (All-contact grand-canonical pressure). *For $T \leq T_*$ the exact hard-sphere density and actual-contact fields have a normally convergent marked real-trajectory pressure on every bounded complex source ball. Its limit is the Hamilton–Boltzmann pressure with collision factor $e^{\Delta p + \psi} - 1$. All fixed source derivatives converge, and the surplus-contact sector vanishes with modulus $\omega_T(\varepsilon)$.*

Proof. Creation trees are absolutely summable by the time simplex and label weights. Theorem 1.3 removes all cyclic genealogies after summation, while Theorem 1.4 gives the source-uniform chronological limit. Collision coarea on the remaining trees produces the Boltzmann collision factor. Normal convergence justifies all fixed source derivatives. $\square$

1.4. Balance-preserving recovery. Let M be a positive Maxwellian reference. Linearize its collision operator as $\mathcal{L}_M$ on the invariant-orthogonal subspace.

Lemma 1.6 (Continuous kinetic Hodge repair). *For every smooth compactly supported balance defect d orthogonal to mass, momentum, and energy, there is a pair (u, Ξ) satisfying*

$$(\partial_t + v \cdot \nabla_x)u - \Delta^* \Xi = d, \quad \|u\|_X + \|\Xi\|_Y \leq C\|d\|_Z,$$

with zero prescribed endpoint moments, pre/post symmetry, and $\Xi \ll A_M$. The right inverse is continuous under velocity truncation and space-time smoothing.

Proof. Solve the nonautonomous linear Boltzmann equation $(\partial_t + v \cdot \nabla_x - \mathcal{L}_M)u = d$ by Duhamel on the invariant-orthogonal subspace, using the classical Maxwellian collision gap and transport energy estimate. Represent $\mathcal{L}_M u$ by its symmetric binary-collision flux to define Ξ . Correct the five endpoint moments by finite smooth homogeneous solutions. This gives the estimate and all symmetries. Since the construction is integral and not cellwise, its norm is uniform under mesh refinement. $\square$

Theorem 1.7 (Grand-canonical and microcanonical joint LDP). *The density-actual-contact pair satisfies a full good short-time LDP under the grand-canonical law, with action*

$$I_0^{\text{gc}}(f_0) + \int \ell(d\Gamma/dA_f) dA_f$$

after imposing the exact weak balance. After the B1 source-dependent shell transfer, the same dynamic action holds with constrained initial rate $I_{0,a}$. Every finite-action pair has a positive smooth tilted recovery.

Proof. Theorem 1.5 on increasing bounded real-source balls gives the upper bound; exponential moment estimates give compact containment. For a smooth positive balanced pair use the exact contact tilt $\psi = \log(d\Gamma/dA_f) - \Delta p$ and the normal pressure derivatives to obtain concentration and the entropy cost. Approximate an arbitrary finite-action pair by truncation and smoothing, add a vanishing Maxwellian background, and use Lemma 1.6 to restore balance exactly; its norm and entropy costs vanish. This proves the lower bound and goodness. B1 transfers every source and recovery law to the primitive shell, changing only the initial rate. $\square$

Theorem 1.8 (Round-twelve B2 closure). *Actual hard-sphere contacts admit a closed trace evolution, a source-uniform all-contact trajectory pressure, and*

the full joint LDP of Theorem 1.7. Recollisions are controlled by summed physical Jacobi transversality, without a false depth-uniform analytic exponent.

Proof. Combine Theorems 1.1, 1.3, 1.5, and 1.7. □

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-twelve dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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